DEVIXO SOLUTIONS

Web Development Internship Program

Professional Personal Portfolio Website

Task 01 | Project Report

Student Name	**Hamza Saeed**
Internship Domain	**Web Development**
Task Number	**01**
Institute	**Government Islamia College, Gujranwala**
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GitHub Repository	**https://github.com/Code-with-Humza/portfolio-website**

1. Objective

The objective of this task was to design and build a modern, professional and fully responsive personal portfolio website using only HTML5 and CSS3. The site had to present my profile, technical skills, education, projects and contact details through a clean interface that works properly on desktop, laptop, tablet and mobile screens.

The task also required me to write all the code myself, without Bootstrap, Tailwind CSS, other UI frameworks or downloaded templates, and to document the work in a GitHub repository with a README file and this report.

2. Problem Statement

A CV lists skills, but it does not show how someone actually builds things. A portfolio website solves that problem by letting visitors see the work, the design thinking and the way a developer organizes code. The challenge in this task was to create such a site from scratch with the following requirements:

- Six required sections: Home, About, Skills, Education, Projects and Contact, plus a navigation bar and a footer.
- A layout that adapts to desktop, laptop, tablet and mobile screens without breaking or scrolling sideways.
- Skill cards, project cards and a contact form interface, all written in plain HTML and CSS.
- Smooth scrolling and subtle hover animations, with a consistent color scheme, readable typography and good spacing.
- Clean, commented code, semantic HTML, meaningful class names and a well-organized project folder.

3. Technologies Used

Technology	How I used it
HTML5	Page structure with semantic elements: header, nav, main, section, article, aside and footer.
CSS3	All styling, including Flexbox, Grid, media queries, transitions, keyframe animations and scroll-driven animations.
JavaScript	A short script (about 20 lines) that highlights the active link in the navigation bar while scrolling. Nothing else depends on it.
Google Fonts	The Manrope typeface for all text.
Font Awesome 6.5.2	An icon font, used only for the GitHub, LinkedIn and email icons.
Git and GitHub	Version control and hosting of the repository.
Browser developer tools	Testing the layout at different screen widths with the device toolbar.

No CSS framework and no website template was used. The layout, components and animations are written by hand in **index.html** and **style.css**.

4. Website Structure

The website is a single page. Every section has its own id, so the navigation links scroll smoothly to the right place.

Section	Id	Purpose
Navigation bar	-	Fixed bar with the brand name, six links and a "Let's Talk" button. On tablet and mobile it turns into a hamburger icon and a sidebar.
Home (hero)	#home	Name, tagline, short introduction, call-to-action buttons and profile photo.
About	#about	Short introduction, academic background, career interests and a quick facts card.
Skills	#skills	Skill cards grouped into Frontend, Tools and Workflow, and Currently Learning.
Education	#education	Degree, institute, study period, coursework and final year project.
Projects	#projects	Three project cards with screenshot, description, technologies and links.
Contact	#contact	Email, phone, location, social links and a contact form interface.
Footer	-	Name, role, social links, copyright and a back-to-top link.

Project folder

```
portfolio-website/
  index.html
  style.css
  README.md
  images/  (profile, icons, skill logos, project screenshots)
  screenshots/  (section and mobile screenshots)
  report/  (this PDF report)
```

5. Design Approach

I designed the website before writing the code. I planned the layout and color scheme at five screen widths, and then built the real pages from that design. The goal was a clean, professional look that still feels a little different from the usual blue-and-white portfolio.

Color scheme

I chose a cool, Scandinavian-style palette with one accent color. A pale blue-grey page background keeps the cards bright, a glacier blue marks interactive elements, and a dark navy gives the hero, footer and sidebar a strong contrast. A small touch of pink is used sparingly as a highlight.

Color	Hex code	Used for
	#F3F7F9	Page background
	#FFFFFF	Cards and navigation bar
	#12232E	Headings and main text
	#5B6F7C	Secondary text
	#DBE5EA	Borders and dividers
	#2B6A8F	Accent: links, icons, buttons
	#0A1B27	Hero, footer and sidebar
	#7BE8FF	Buttons and labels on dark areas
	#FF3D8B	Rare highlight (timeline dot, status dot)

Typography

All text uses the Manrope font in weights from regular to extra bold. Headings are bold and dark, descriptions use a softer grey, and line height is kept generous (about 1.6) so paragraphs stay easy to read.

Layout and spacing

Content sits in a centered container with a maximum width of 75rem. Cards share one style: white background, a thin 1px border and rounded corners, with no heavy shadows. Spacing between sections and inside cards is consistent across the whole page.

Responsive design

The layout changes at several breakpoints (1366px, 1100px, 820px, 560px and 340px). On large screens sections use two or three columns. On tablets the navigation links are replaced by a hamburger icon, and cards drop to two columns. On phones everything becomes a single column, and the sidebar becomes narrower (300px on tablet, 220px on phone and 172px on very small phones).

Motion

Animations are short (about 0.2 to 0.5 seconds), smooth and never loop. They only change opacity and position, and they are switched off for visitors who have asked their device to reduce motion.

6. Website Features

- **Navigation bar:** fixed at the top, with animated underlines on hover, a highlighted active link and a soft shadow that fades in while scrolling.
- **Hero section:** dark background, name, tagline, short introduction, two call-to-action buttons and a profile photo, all entering with a short staggered animation.
- **About section:** short introduction, academic background and career interests cards, and a quick facts card with a Download Resume button.
- **Skills section:** three skill cards with icons and one-line descriptions. Rows react on hover and the cards appear as they scroll into view.

- **Education section:** a timeline with a line that draws itself and a dot that pops in, plus study details, coursework and the final year project.
- **Projects section:** three project cards with a screenshot, description, technology tags, and live demo and GitHub links. Cards lift and the image zooms slightly on hover.
- **Contact section:** contact details, social links and a form with name, email, subject, message and a send button. The form is an interface only and has no backend.
- **Footer:** name, role, social icons, copyright line and a back-to-top link.
- **Hamburger menu and sidebar:** on tablet and mobile the navigation moves into a sidebar with a profile photo, links, a Let's Talk button, social icons and the copyright line. It is built with pure CSS using the :target selector.
- **Smooth scrolling and hover effects:** smooth scrolling between sections, and consistent hover effects on links, buttons, cards and icons.
- **Responsive layout:** works on desktop, laptop, tablet and mobile without horizontal scrolling.

7. Implementation Steps

Step 1: Planning the content. I wrote the text for every section first: introduction, skills, education, projects and contact details. This meant I never had to invent copy while coding.

Step 2: Designing the look. I planned the layout, color palette and typography, and checked how each section should look at desktop, laptop, tablet and mobile widths.

Step 3: Writing the HTML structure. I built the full page skeleton with semantic elements, one section per required part, meaningful class names and working anchor links.

Step 4: Setting up the base CSS. I added a reset, the font, page colors, smooth scrolling and the shared card and button styles that the rest of the page reuses.

Step 5: Styling the navbar and hero. I styled the fixed navigation bar, the animated underline links, the dark hero section and the call-to-action buttons.

Step 6: Styling the remaining sections. I built the About, Skills, Education, Projects, Contact and Footer sections using Flexbox and Grid.

Step 7: Adding animations. I added the hero entrance animation, scroll-triggered reveals, the timeline drawing effect and hover effects, keeping everything short and subtle.

Step 8: Making it responsive. I replaced fixed sizes with flexible ones, added media queries for each screen size, and built the hamburger icon and sidebar with CSS only.

Step 9: Testing. I tested the site at different widths with the browser device toolbar, checked that every link works and that no page scrolls sideways.

Step 10: Documentation and upload. I took screenshots, wrote the README and this report, and uploaded everything to a GitHub repository.

8. Code Screenshots

The screenshots below show the main parts of the code with a short explanation of each. They are shortened excerpts of the real files.

```html
<!-- Top navigation bar -->
<nav class="navbar">
  <a href="#home" class="nav-brand">Hamza Saeed</a>
  <ul class="header-nav-links">
    <li><a href="#home">Home</a></li>
    <li><a href="#about">About</a></li>
    <li><a href="#skills">Skills</a></li>
    <li><a href="#education">Education</a></li>
    <li><a href="#projects">Projects</a></li>
    <li><a href="#contact">Contact</a></li>
  </ul>
  <a href="#contact" class="lets-talk">Let's Talk</a>
  <a href="#menu" class="hamburger" aria-label="Open menu">
    <span></span><span></span><span></span>
  </a>
</nav>

<!-- Sidebar for tablet and mobile (opens with :target) -->
<aside class="sidebar" id="menu" aria-label="Site menu">
  <a href="#close" class="sidebar-scrim" tabindex="-1"></a>
  <div class="sidebar-panel">
    <a href="#close" class="sidebar-close" aria-label="Close menu">&#10005;</a>
    ...
  </div>
</aside>
```

Figure 1: HTML for the navigation bar and sidebar

The navbar contains the brand link, the six section links, the Let's Talk button and a hamburger icon. The hamburger is a normal link to #menu. The sidebar below it is hidden until that link is clicked, and its close button and background link to #close. All links are real anchors, so the menu works without JavaScript.

```html
<!-- Hero -->
<section class="hero-section" id="home">
  <div class="hero-text">
    <div class="intro-line">FRONTEND WEB DEVELOPER</div>
    <h1 class="hero-name">Hamza Saeed</h1>
    <div class="tagline">I build clean, responsive websites.</div>
    <p class="description">
      IT student from Gujranwala working with HTML, CSS and JavaScript,
      and learning React. I'm looking for a frontend internship where I
      can build real products.
    </p>
    <a href="#contact" class="hero-link lets-talk-button">Let's Talk</a>
    <a href="#projects" class="hero-link view-projects-button">View Projects</a>
  </div>
  <div class="hero-portrait">
    <img src="images/profile.png" alt="Portrait of Hamza Saeed"
         width="400" height="480" class="hero-profile" />
  </div>
</section>
```

Figure 2: HTML for the hero section

The hero uses a section with the id home. It holds the text block (label, h1 name, tagline, description and two buttons) and a wrapper around the portrait. The image has width and height attributes and descriptive alt text, so the page does not jump while loading.

```css
*,
*::before,
*::after {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

html {
  scroll-padding-top: 5rem;
}

@media (prefers-reduced-motion: no-preference) {
  html {
    scroll-behavior: smooth;
  }
}

body {
  background-color: #f3f7f9;
  color: #12232e;
  font-family: "Manrope", system-ui, sans-serif;
  line-height: 1.5;
}

.about-card,
.fact-card,
.skill-card,
.education-card,
.first-project,
.user-contact-form {
  background-color: #fff;
  border: 1px solid #dbe5ea;
  border-radius: 0.8rem;
}
```

Figure 3: Base CSS: reset, smooth scrolling and shared card style

The reset removes default spacing and uses border-box sizing. scroll-padding-top keeps section titles from hiding under the fixed navbar, and smooth scrolling is only turned on for visitors who have not asked for reduced motion. One grouped rule gives every card the same white background, border and rounded corners, which avoids repeating the same code.

```css
.sidebar {
  position: fixed;
  inset: 0;
  z-index: 20;
  visibility: hidden;
  transition: visibility 0s 0.3s;
}

.sidebar:target {
  visibility: visible;
  transition-delay: 0s;
}

.sidebar-panel {
  position: absolute;
  top: 0;
  right: 0;
  width: 18.75rem;
  height: 100%;
  background-color: #0a1b27;
  translate: 100% 0;
  transition: translate 0.3s cubic-bezier(0.2, 0.7, 0.2, 1);
}

.sidebar:target .sidebar-panel {
  translate: 0;
}
```

Figure 4: CSS for the sidebar

The sidebar stays hidden with visibility until it becomes the :target of the link. The panel then slides in from the right with the translate property. Because visibility switches back with a short delay, the panel can slide out smoothly when it closes, and hidden links cannot be tabbed to.

```css
.hero-text > * {
  animation: rise 0.5s cubic-bezier(0.2, 0.7, 0.2, 1) both;
}

.hero-text > :nth-child(2) {
  animation-delay: 80ms;
}

@keyframes rise {
  from { opacity: 0; transform: translateY(12px); }
  to   { opacity: 1; transform: none; }
}

@supports (animation-timeline: view()) {
  .reveal,
  .skill-card,
  .first-project {
    animation: reveal linear both;
    animation-timeline: view();
    animation-range: entry 0% entry 40%;
  }
}
```

Figure 5: CSS for the animations

Each child of the hero text uses the same rise animation, with a slightly longer delay for each, so the content appears in reading order. The scroll reveal uses animation-timeline: view(), which plays the animation as an element enters the screen. It sits inside @supports, so browsers without that feature simply show the content normally.

```css
@media (max-width: 820px) {
  .header-nav-links {
    display: none;
  }

  .hamburger {
    display: flex;
  }

  .hero-section {
    flex-direction: column;
    align-items: flex-start;
  }

  .skills-card-container,
  .project-cards-container {
    grid-template-columns: 1fr 1fr;
  }
}

@media (max-width: 560px) {
  .navbar .lets-talk {
    display: none;
  }

  .skills-card-container,
  .project-cards-container {
    grid-template-columns: 1fr;
  }
}
```

Figure 6: CSS media queries

At 820px and below, the navigation links are hidden and the hamburger appears, the hero stacks into one column and the card grids drop to two columns. At 560px and below, the Let's Talk button moves into the sidebar and the grids become a single column.

9. Desktop Screenshots (Output)

These screenshots were taken at desktop and laptop widths and show each section of the finished website.

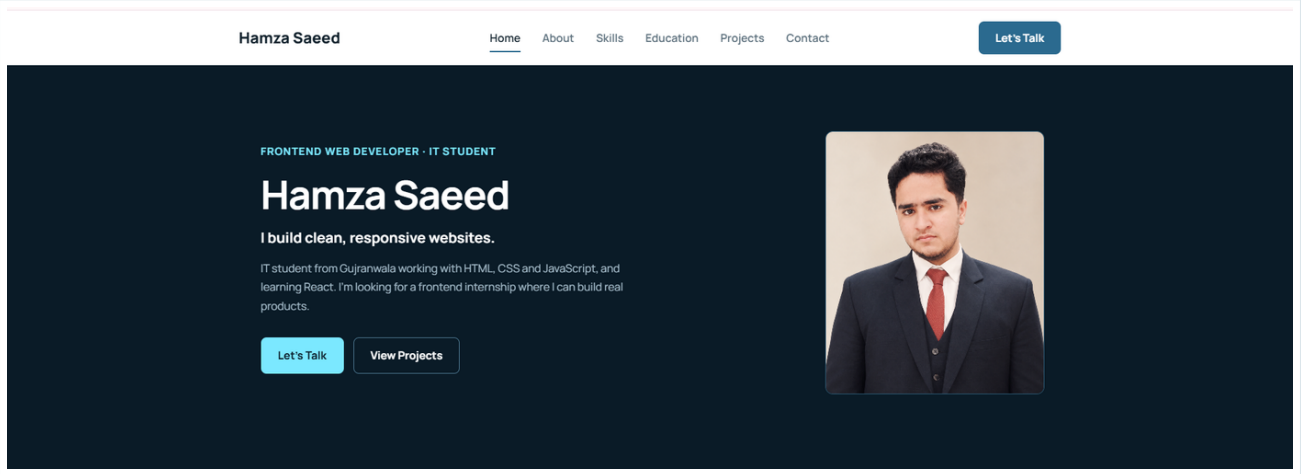


Figure 7: Home / hero section

The white navigation bar sits above the dark hero. The active link (Home) has an underline, and the Let's Talk button stands out in glacier blue. The hero shows the label, my name, the tagline, a short introduction, two buttons and my profile photo.

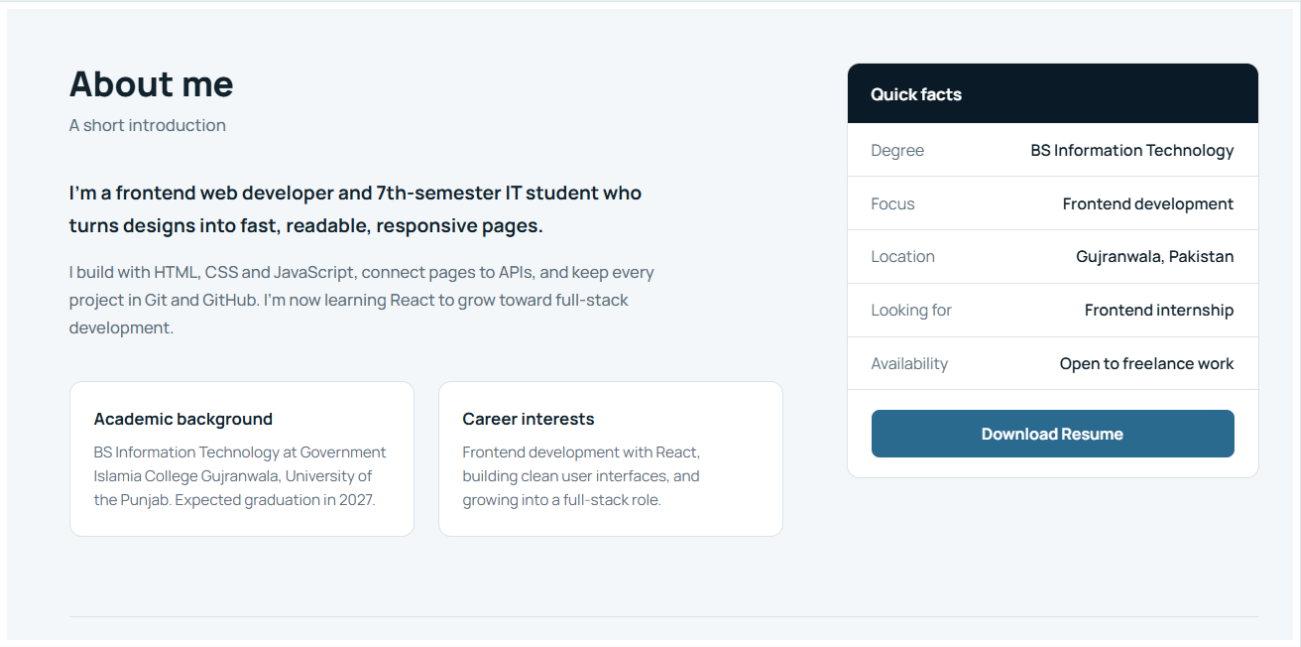


Figure 8: About section

The left side has the section title, a lead sentence, a short paragraph and two cards for academic background and career interests. The right side has the quick facts card with a dark header, five fact rows and a Download Resume button.

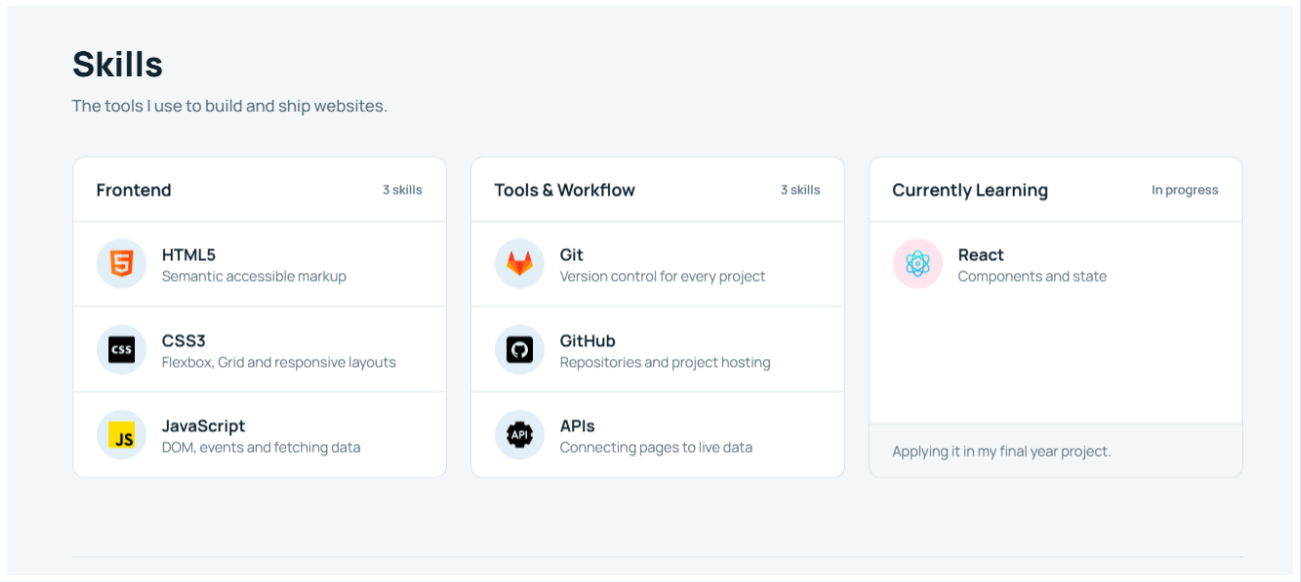


Figure 9: Skills section

Skills are grouped into three cards: Frontend (HTML5, CSS3, JavaScript), Tools and Workflow (Git, GitHub, APIs) and Currently Learning (React). Each row has an icon, a name and a one-line description, and the cards stay the same height.

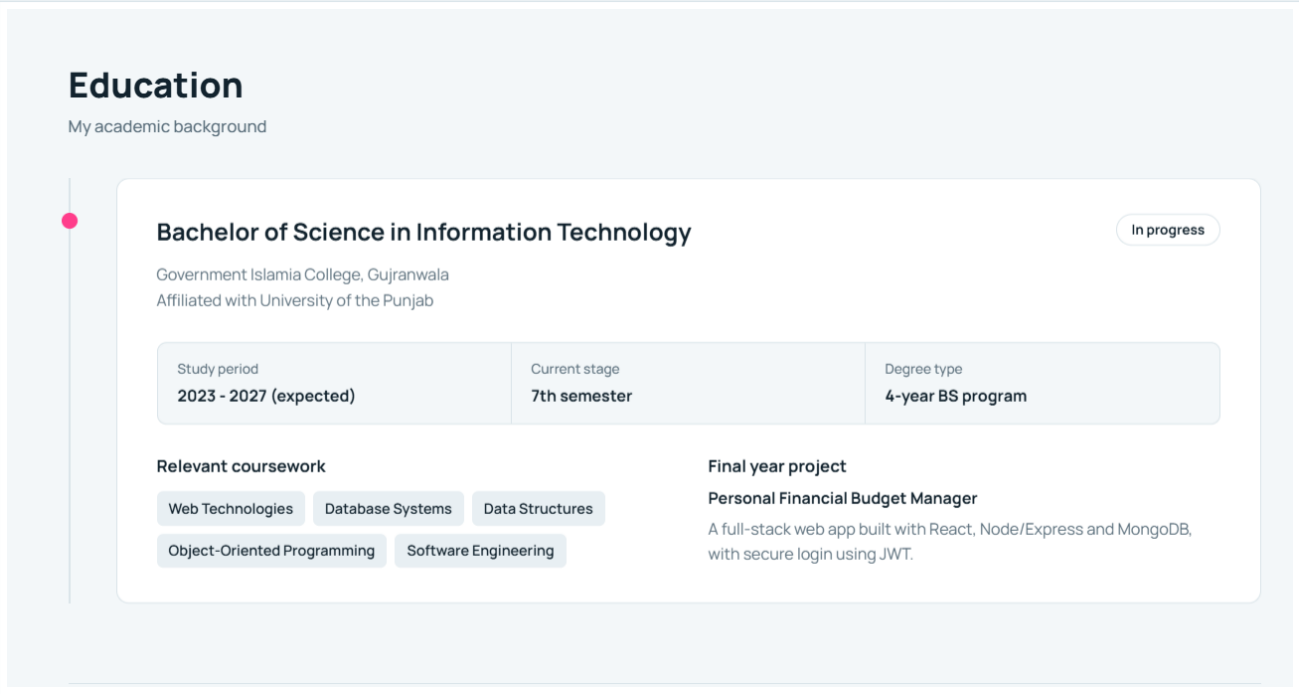


Figure 10: Education section

A vertical timeline with a pink dot leads to the education card. The card shows the degree, the institute and university, an In progress tag, a three-part strip with study period, current stage and degree type, the relevant coursework tags and my final year project.

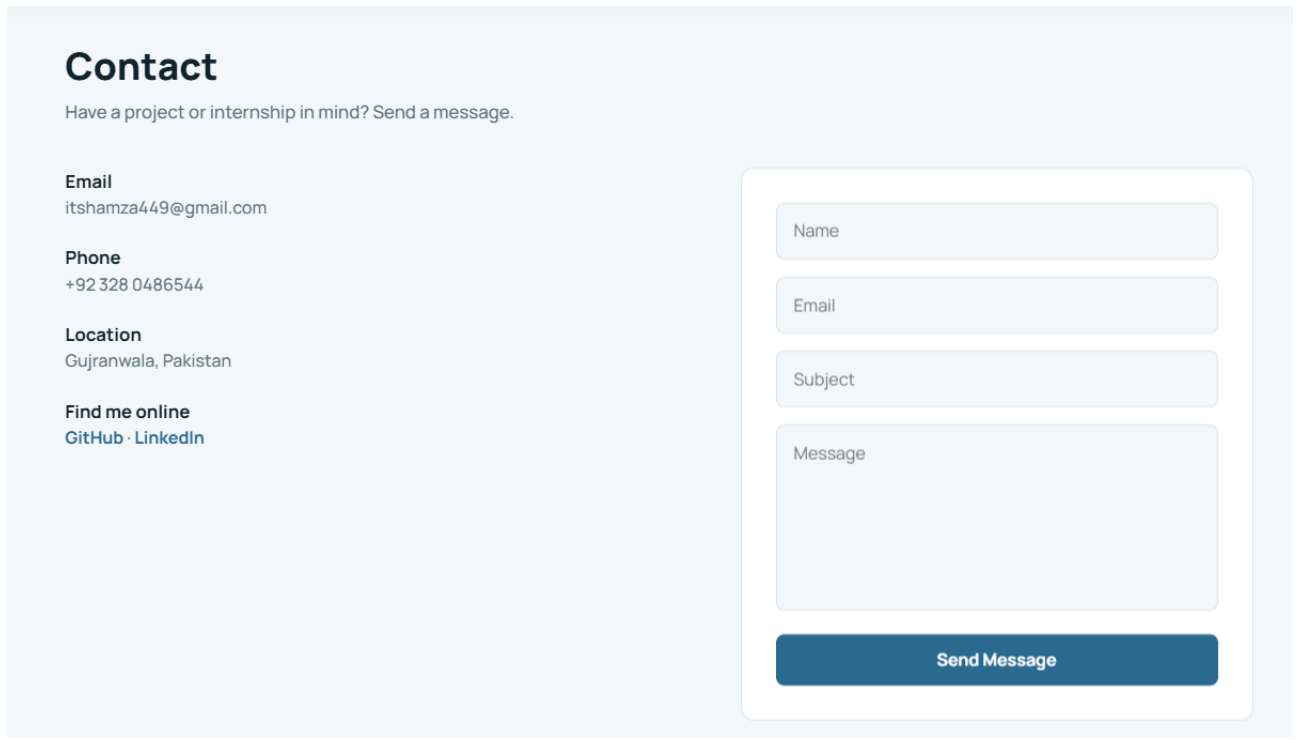


Figure 11: Contact section

The left column lists email, phone, location and social links. The right column is the contact form with name, email, subject and message fields and a Send Message button. The form is only an interface, as the task allows.

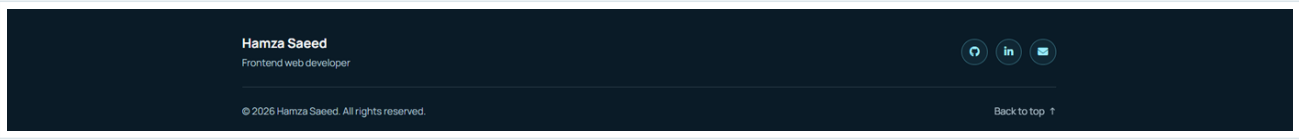


Figure 12: Footer

The dark footer shows my name and role, round social icons for GitHub, LinkedIn and email, the copyright line and a Back to top link.

10. Project Screenshots

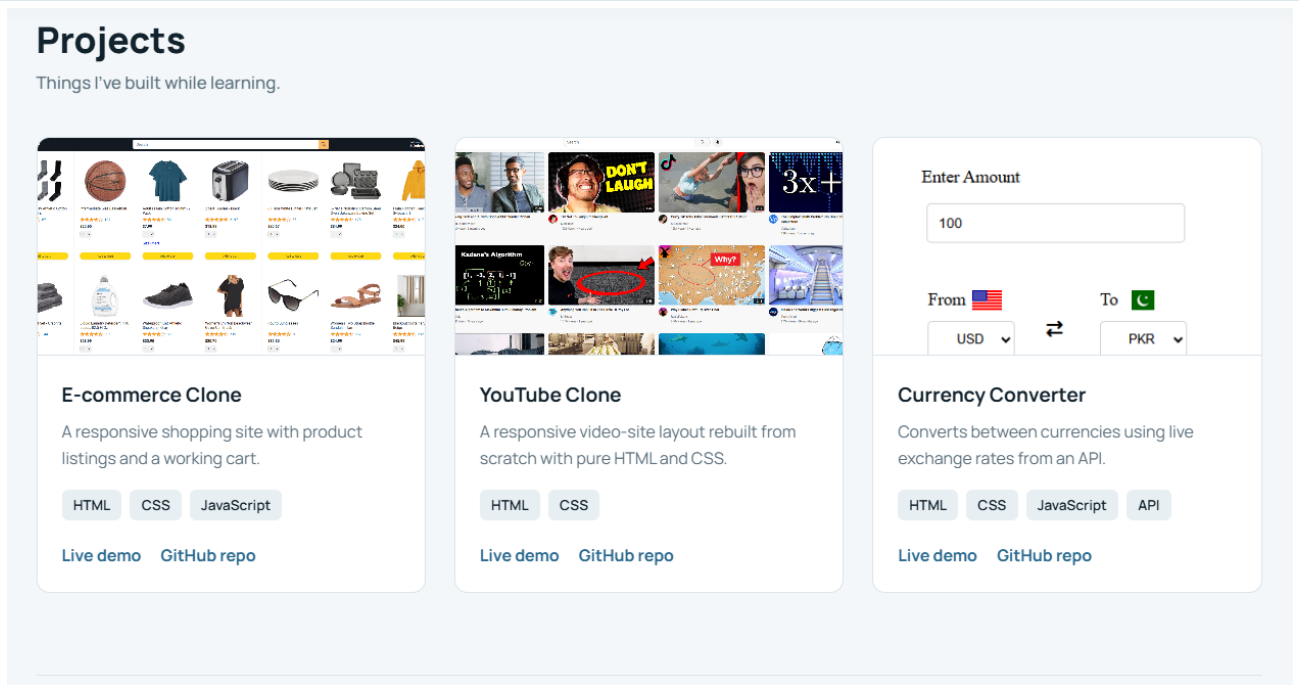


Figure 13: Projects section

Three project cards show my work, each with a screenshot, a short description, the technologies used and two links (Live demo and GitHub repo). All cards share the same structure, so the row looks organized.

Project	Technologies	Description
E-commerce Clone	HTML, CSS, JavaScript	A responsive shopping site with product listings and a working cart.
YouTube Clone	HTML, CSS	A responsive video-site layout rebuilt from scratch with pure HTML and CSS.
Currency Converter	HTML, CSS, JavaScript, API	Converts between currencies using live exchange rates from an API.

11. Mobile Screenshots

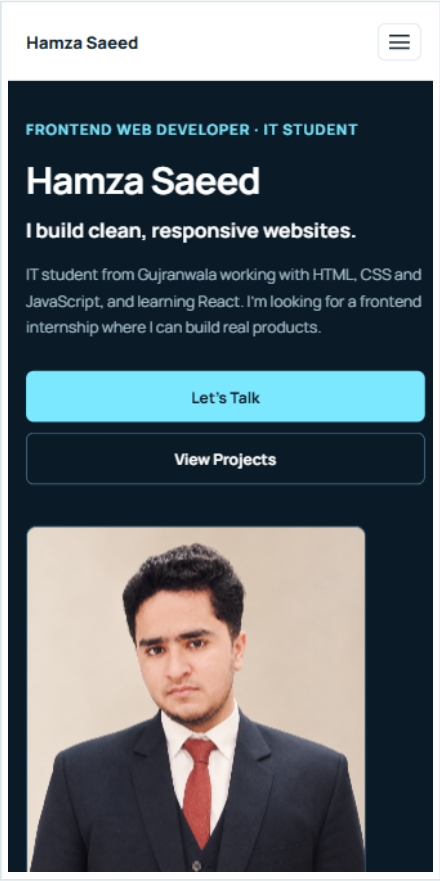


Figure 14: Mobile view

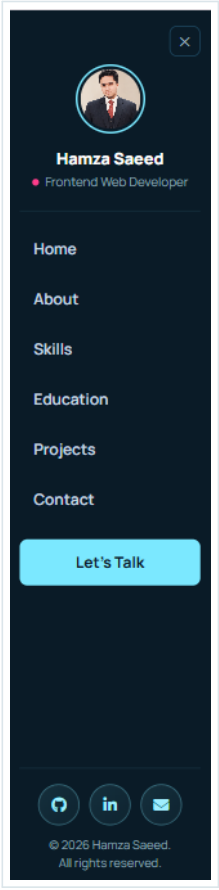


Figure 15: Sidebar menu open

On a phone the navigation links are replaced by a hamburger icon, the hero becomes a single column and the buttons stretch across the screen. Tapping the hamburger opens the sidebar, which has a round profile photo, my name and role, the section links, a Let's Talk button, social icons and the copyright line. The sidebar is narrower on small screens so it does not cover most of the page.

12. Challenges Faced

- **A sidebar without JavaScript.** A mobile menu normally needs JavaScript to open and close. To stay within HTML and CSS, I used the :target selector: the hamburger links to the sidebar, and the links inside it point to other sections, so the menu closes by itself when one is clicked. The side effect is that the address bar shows a hash when the menu is open, which I accepted.
- **Hover effects fighting with animations.** Some cards have both a scroll animation and a hover lift. Because the animation keeps its final transform value, the hover lift stopped working. I fixed it by using the separate translate property for the hover movement.
- **Turning a laptop layout into a responsive one.** My first layout used fixed widths and heights in rem units, which looked right on a laptop but not on smaller screens. I replaced them with Flexbox, Grid, maximum widths and media queries, and fixed the horizontal scrolling that the slide-in animation could cause by clipping overflow on that section.
- **Browser support for scroll animations.** Scroll-driven animations only work in some browsers. I placed them inside @supports, so other browsers show the same content without the entrance effect

instead of hiding it.

- **Keeping the design simple.** It was tempting to add many effects. I limited myself to short, subtle animations and one accent color, so the website stays clean and professional.

13. Conclusion

This task helped me move from writing separate HTML and CSS examples to building a complete website with a consistent design. I now understand much better how to structure a page with semantic HTML, how Flexbox and Grid work together, and how media queries make one layout work on every screen size.

Building the hamburger menu and the animations without JavaScript taught me how much CSS can do on its own, and why details such as focus styles, reduced-motion support and fallbacks matter. The finished portfolio meets the task requirements, and I plan to keep updating it with new projects as I continue learning React and move toward full-stack development.

14. GitHub Repository Link

The complete project, including the source code, images, screenshots, README file and this report, is available here:

https://github.com/Code-with-Humza/portfolio-website